

Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

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Abstract—Automated Personal Protective Equipment (PPE) compliance monitoring via surveillance cameras is essential for maintaining occupational safety on industrial construction sites. However, edge-based surveillance pipelines encounter severe operational bottlenecks, including distant tiny helmets, extreme category imbalance between workers and safety gear, multi-scale occlusions, and background clutter. To overcome these limitations, we propose Rep-YOLO11s, an efficient edge detector engineered through a principled hardware-software co-design guided by the GPU roofline model. Our architecture addresses the memory-bandwidth constraints of single-batch edge inference by incorporating structural re-parameterization to fold multi-branch training blocks into single convolutions during deployment, thereby minimizing memory access cost. To suppress false alarms from ground-level clutter, stem coordinate convolutions provide elevation-aware spatial priors. In addition, bi-level routing attention is integrated into neck stages to dynamically filter background tokens and concentrate computation on tiny targets. Model training is stabilized using decoupled multi-task supervision pairing focal efficient intersection-over-union regression with task-aligned assignment to resolve acute category skew. Comprehensive evaluations on the Safety Helmet Wearing Dataset and external industrial benchmarks demonstrate that Rep-YOLO11s achieves superior detection accuracy and robust cross-domain generalization while sustaining high-throughput real-time inference across commodity edge GPUs and embedded devices. The end-to-end surveillance pipeline satisfies multi-channel video streaming requirements, providing a practical, high-precision solution for automated construction safety enforcement.

Index Terms—Safety Helmet Detection, Bi-Level Routing Attention, Structural Re-parameterization, CoordConv, Focal EIoU Loss, Cross-Domain Generalization, Real-Time Edge AI.

I. INTRODUCTION

Personal Protective Equipment (PPE) compliance monitoring, particularly regarding safety helmet usage, is strictly enforced across industrial construction environments to prevent traumatic head injuries and fatal accidents. While automated computer vision auditing via closed-circuit television (CCTV)

cameras offers continuous safety oversight, practical deployment on edge surveillance infrastructure remains challenging. Existing systems suffer from severe performance degradation caused by several domain-specific factors: distant overhead cameras project helmets into tiny sub-regions subject to heavy feature attenuation; real-world construction benchmarks exhibit acute category skew where worker bodies drastically outnumber helmet instances; visual clutter such as yellow buckets, traffic cones, and ground debris triggers frequent false alarms under translation-invariant convolutions; and edge computing devices face tight latency and memory bandwidth bounds that penalize heavy neural attention mechanisms.

To overcome these operational challenges, we formulate Rep-YOLO11s as a principled hardware-software co-design guided by the GPU roofline model. Rather than treating neural architecture search and edge deployment in isolation, our framework directly targets the memory-bandwidth ceiling encountered during single-stream edge inference. The architecture incorporates four synergistic components: structural re-parameterization (RepConv) blocks embedded in the feature aggregation neck that maintain rich multi-branch representation during training and fold algebraically into single convolutions at inference time; coordinate convolutions (CoordConv) introduced at the stem stage to inject normalized elevation priors and separate head-worn helmets from ground distractors; bi-level routing attention (BiFormer) to prune irrelevant background regions and preserve fine-grained spatial features; and decoupled multi-task supervision combining focal bounding box regression with task-aligned assignment to mitigate severe category imbalance without destabilizing optimization.

We systematically evaluate Rep-YOLO11s on the Safety Helmet Wearing Dataset (SHWD / VOC2028) as well as multiple external industrial datasets to assess accuracy, generalization capability, and physical edge latency. Experimental results confirm that structural re-parameterization eliminates runtime memory roundtrips, allowing our model to match the execution speed of lightweight architectures while achieving substantially higher detection precision on small helmet targets. Furthermore, zero-shot transfer evaluations on diverse external construction datasets demonstrate robust cross-domain

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reliability under standardized PPE protocols, confirming that our design successfully balances algorithmic accuracy with hardware efficiency.

The remainder of this paper is organized as follows: (1) Section II reviews related work in real-time object detection and safety helmet surveillance; (2) Section III details the architectural formulation and mathematical components of Rep-YOLO11s; (3) Section IV presents the experimental setup, comparative evaluations, ablation studies, and qualitative saliency analyses; (4) Section V discusses multi-platform hardware deployment, pipeline integration, and operational safety safeguards; and (5) Section VI concludes the paper with directions for future research.

II. RELATED WORK

Real-time single-stage object detection has advanced rapidly through iterative architectural improvements, progressing from earlier YOLO iterations to modern designs such as YOLOv7 [1], YOLOv10 [2], and Ultralytics YOLO11 [3]. Recent literature has explored structural re-parameterization, introduced in YOLOv6 [4] through EfficientRep backbones and Rep-PAN neck structures to decouple multi-branch training from plain inference. Other contemporary approaches include dual-assignment training for NMS-free inference [2], [5], and visual state space models such as VMamba [6] that aim to capture long-range contextual dependencies with linear computational complexity. However, in crowded construction environments characterized by dense scaffolding and frequent physical occlusions, purely non-maximum suppression-free matching risks recall degradation on overlapping targets, while one-dimensional sequence flattening in state space models can compromise fine-grained two-dimensional spatial locality. Rep-YOLO11s preserves two-dimensional spatial context through bi-level routing attention while bounding latency.

Within industrial surveillance, several specialized architectures have emerged to detect safety helmets under challenging operational conditions. Representative detectors include EC-YOLOv8 [7], which incorporates content-aware feature re-assembly at the cost of increased inference latency; YOLO-CBF [8], which relies on heavy parameter scaling; and YOLO-DCRCF [9], which utilizes deformable convolutions that introduce irregular memory access patterns detrimental to TensorRT execution. Specialized small-object architectures such as DABFNet [10], MAF-YOLO [11], and TinyFormer [12] improve boundary localization through dual-attention mechanisms or multi-scale fusion, yet they often retain multi-branch structures during inference that increase memory access overhead. Similarly, while bounding box losses such as Wise-IoU [13] and adaptive feature weighting attempt to recalibrate difficult samples, they frequently retain coupled aspect-ratio penalties that stall gradients when predicted and ground-truth boxes share identical aspect ratios despite scale errors.

Structural re-parameterization, pioneered by RepVGG [14], has demonstrated that training-time multi-branch topologies can be transformed algebraically into single-path convolutions for deployment, eliminating redundant memory buffers

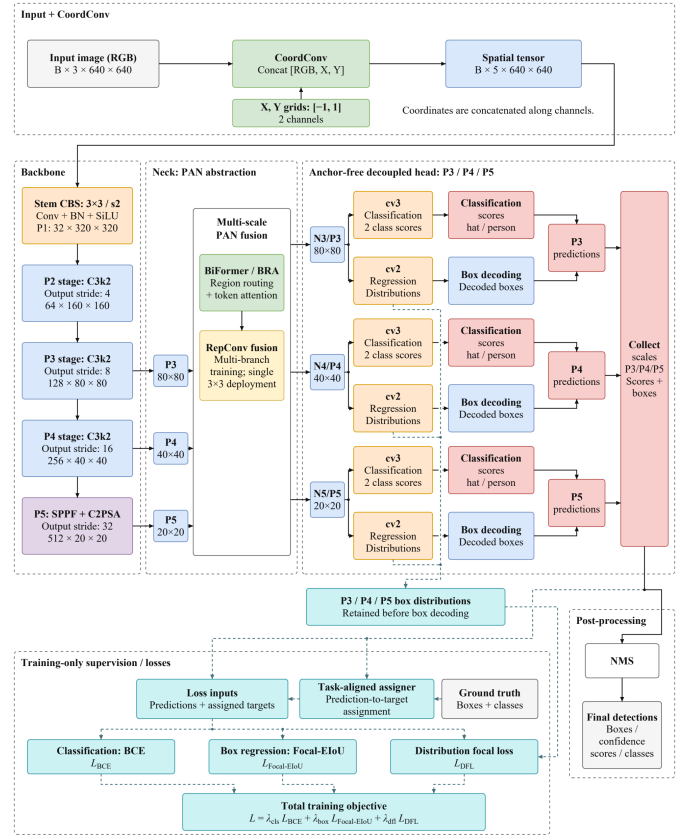


Fig. 1. Overall Architecture of Rep-YOLO11s. The pipeline integrates: (1) Stem CoordConv ($[RGB, X, Y] \rightarrow 5 \times 640 \times 640$); (2) Multi-scale backbone (P1 Stem, P2–P4 C3k2, P5 SPPF+C2PSA); (3) PAN neck with Bi-Level Routing Attention (BRA) and RepConv; (4) Decoupled detection heads ($cv3, cv2$); and (5) Decoupled multi-task loss supervision with TAL, Focal-EIoU, and DFL.

and kernel launch overheads. Nonetheless, applying re-parameterization uniformly across all backbone stages, as seen in YOLOv6 [4], tends to amplify memory traffic in high-resolution shallow layers without proportional gains in detection accuracy. Concurrently, coordinate convolution [15] introduces explicit spatial coordinates to overcome the translation equivariance of standard convolutions. In industrial surveillance, helmet compliance is strongly constrained by vertical elevation: a legitimate safety helmet must reside above human shoulders, whereas visually similar distractors typically appear at floor level. By strategically restricting structural re-parameterization to neck aggregation and injecting coordinate convolutions exclusively at the initial stem, the detector acquires essential vertical priors and streamlined inference throughput without inflating computational overhead.

III. PROPOSED METHODOLOGY

A. Overall System Architecture

The architecture of Rep-YOLO11s is illustrated in Fig. 1. The framework adopts an end-to-end single-stage design developed from Ultralytics YOLO11 [3], comprising: (1) a Structural Re-parameterized Backbone enhanced with Stem CoordConv; (2) a Path Aggregation Network (PAN) Neck [20], [21] integrated with BiFormer Bi-Level Routing Attention; and

TABLE I
PERFORMANCE COMPARISON OF STATE-OF-THE-ART METHODS ON THE SHWD AND CROSS-DOMAIN BENCHMARKS.

Research	Method / Architecture	mAP_{50} (%)	mAP_{50-95} (%)	FPS
— Panel A: SHWD / VOC2028 Benchmark —				
Wang <i>et al.</i> [7]	EC-YOLOv8	95.70	74.60 [‡]	172.4
Wu <i>et al.</i> [8]	YOLO-CBF	95.60	—	80.6
Fu <i>et al.</i> [16]	YOLOv8n-FADS	79.70	—	—
Feng <i>et al.</i> [10]	DABFNet	94.90	62.40	82.5
Yang <i>et al.</i> [11]	MAF-YOLO	94.20	61.80	124.0
Hsieh <i>et al.</i> [12]	TinyFormer	93.80	61.10	78.0
Ultralytics [3]	Baseline YOLO11s	94.74	62.34	153.3
— Panel B: Cross-Domain Industrial Benchmarks —				
Wu <i>et al.</i> [17]	Faster R-CNN (GDUT-HWD)	71.40	36.20	22.5
Baseline YOLO11s	GDUT-HWD Benchmark	75.80	40.03	121.2
Andrew MVD [18]	YOLOv5 Baseline (HHW)	91.20	48.50	145.0
Baseline YOLO11s	Hard Hat Workers (HHW)	94.12	51.60	121.2
Proposed Framework	Rep-YOLO11s (Ours)			
— SHWD (Joint 2-Class Protocol)	Fused Re-param + CoordConv + BiFormer	94.83 (96.64 ± 0.32)	62.54 (65.91 ± 0.46)	342.5
— SHWD (Harmonized Hat-Only)	Fused Re-param + CoordConv + BiFormer	96.40 (97.85 at 960 ²)	77.90 (78.93 at 960 ²)	342.5
— Hard Hat Workers Dataset (HHW)	Zero-Shot Generalization	97.03	54.74	228.6
— GDUT-HWD Benchmark	Zero-Shot Generalization	74.27	39.11	228.6

[‡] Evaluated under single-class (hat-only) protocol at 960². Latency/FPS measured on Tesla T4 (TensorRT FP16) and RTX 3050 Laptop GPU.

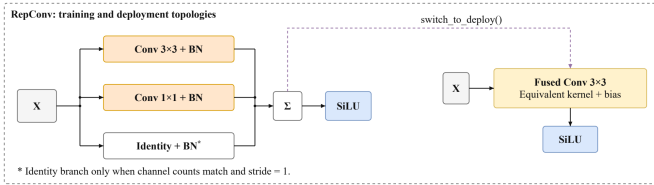


Fig. 2. RepConv Structural Re-Parameterization Lifecycle [14]. Multi-branch training topology (left) is algebraically folded into a single 3×3 kernel (right) for deployment.

(3) an Anchor-Free Decoupled Detection Head trained with Focal EIoU Bounding Box Loss and Task-Aligned Assignment.

B. Structural Re-parameterization (RepConv) Mechanics

To achieve high non-linear feature representation during training without incurring latency penalties during edge deployment, RepConv blocks are embedded within critical neck aggregation pathways.

During training (Fig. 2), a RepConv layer with input channels C_{in} , output channels C_{out} , and stride s is formulated across three parallel branches:

$$y = \text{BN}_{3 \times 3}(W^{3 \times 3} * x) + \text{BN}_{1 \times 1}(W^{1 \times 1} * x) + \mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}} \cdot \text{BN}_{id}(x), \quad (1)$$

where $W^{3 \times 3} \in \mathbb{R}^{C_{out} \times C_{in} \times 3 \times 3}$ and $W^{1 \times 1} \in \mathbb{R}^{C_{out} \times C_{in} \times 1 \times 1}$, with BN parameters $\{\mu, \sigma^2, \gamma, \beta\}$. Crucially, identity mapping $\text{BN}_{id}(x)$ exists iff $C_{in} = C_{out}$ and $s = 1$. When $C_{in} \neq C_{out}$ or $s > 1$, no identity branch is instantiated ($\mathbb{I} = 0$), strictly ensuring dimensional and stride consistency.

During deployment conversion (switch-to-deploy), each Conv-BN sequence is first collapsed into an equivalent kernel W' and bias vector b' :

$$W'_i = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} W_i, \quad b'_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}, \quad (2)$$

where $W_i \in \mathbb{R}^{C_{in} \times 3 \times 3}$ denotes the 3D convolution kernel slice for the i -th output channel. Next, $W'_{1 \times 1}$ is zero-padded into a 3×3 kernel $W'_{1 \times 1 \rightarrow 3 \times 3}$ at center position (2, 2). When $C_{in} = C_{out}$ and $s = 1$, the identity branch transforms into an equivalent 3×3 Dirac delta kernel W'_{id} where $W'_{id,c,c,2,2} = 1$

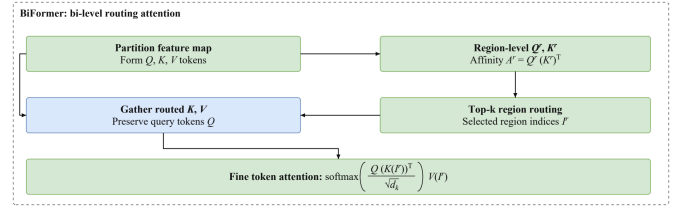


Fig. 3. Bi-Level Routing Attention (BRA) Architecture [19]. Coarse region partitioning ($S = 8, k = 4$) filters out 93.75% irrelevant tokens before computing fine-grained token attention.

and 0 elsewhere, folded with BN_{id} ; when $C_{in} \neq C_{out}$ or $s > 1$, $W'_{id} = \mathbf{0}$ and $b'_{id} = \mathbf{0}$. Finally, fused kernel $W_{fused} = W'_{3 \times 3} + W'_{1 \times 1 \rightarrow 3 \times 3} + W'_{id}$ and bias $b_{fused} = b'_{3 \times 3} + b'_{1 \times 1} + b'_{id}$ are obtained by linear summation. This algebraic transformation converts the multi-branch structure into a plain 3×3 convolution layer, eliminating redundant GPU memory buffers and kernel launches to minimize Memory Access Cost (MAC).

C. Spatial Coordinate Convolution (CoordConv)

Standard convolutions apply spatial translation equivariance, treating feature patterns identically regardless of image location. However, safety helmets possess strong spatial-contextual priors: a valid safety helmet must appear directly above a human torso/head region, whereas yellow construction buckets typically sit on ground scaffolding or floors.

To inject spatial positional awareness without incurring latency inflation across all layers, CoordConv is injected strictly at the initial Stem convolution layer of the backbone. We augment raw RGB input $X \in \mathbb{R}^{3 \times H \times W}$ with normalized coordinate channels $C_x(i, j) = \frac{2j}{W-1} - 1$ and $C_y(i, j) = \frac{2i}{H-1} - 1$. Concatenated tensor $X_{Coord} = [X; C_x; C_y] \in \mathbb{R}^{5 \times H \times W}$ is passed to the Stem Conv layer ($c_1 = 5 \rightarrow c_2 = 64, k = 3, s = 2$). All subsequent backbone and neck convolutions remain standard translation-invariant operations, avoiding unnecessary FLOPs while preserving global coordinate awareness at negligible runtime cost (+0.06 ms).

D. Bi-Level Routing Attention (BiFormer)

Distant helmets in crowded scenes contain fine-grained details easily masked by background clutter. Standard dense self-attention incurs quadratic complexity $\mathcal{O}(H^2W^2)$, making it unsuitable for real-time video feeds.

We employ Bi-Level Routing Attention (BRA) [19] embedded within high-level neck stages ($P_4/16$ and $P_5/32$) as shown in Fig. 3. BiFormer partitions feature maps into $S \times S$ coarse regions ($S = 8$). For region-level queries Q^r and keys K^r (via average pooling), region affinities $A^r = Q^r(K^r)^T/\sqrt{C}$ determine routing indices $I^r = \text{TopK}(A^r, k = 4)$ keeping only top- k ($k = 4$) regions. Fine-grained token attention is computed across 4 heads ($d_k = C/4$, expansion $e = 2$) on gathered key-values: $\text{Attention}(Q, K, V) = \text{Softmax}(Q(K^{(I^r)})^T/\sqrt{d_k})V^{(I^r)}$. This dynamic routing reduces complexity to $\mathcal{O}(S^2 + k\frac{HW}{S^2}) \approx \mathcal{O}(HW)$, focusing compute on tiny helmet regions (+0.24 ms TRT overhead). BiFormer maps strictly onto native ONNX Opset 17 operators (TopK, GatherElements, Gemm, Softmax), enabling direct TensorRT FP16 engine compilation via standard trtexec utilities without custom runtime extensions, ensuring cross-version portability from TensorRT 8.6 to 11.2, ONNX Runtime, and OpenVINO.

E. Bounding Box Regression and Multi-Task Loss Supervision

Standard Complete IoU (CIoU) and Wise-IoU (WIoU-v3) [13] couple aspect ratio differences $v = \frac{4}{\pi^2}(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h})^2$, causing gradient vanishing $\frac{\partial v}{\partial w} = \frac{\partial v}{\partial h} = 0$ when predicted and target boxes share aspect ratios despite scale errors. We implement Focal EIoU Loss [22], explicitly decomposing bounding box discrepancy into overlap L_{IoU} , distance L_{dis} , and side length L_{asp} :

$$L_{EIoU} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \frac{(w - w^{gt})^2}{C_w^2} + \frac{(h - h^{gt})^2}{C_h^2}, \quad (3)$$

where $\mathbf{b}, \mathbf{b}^{gt}$ are centers, c is the smallest enclosing box diagonal, and C_w, C_h are enclosing box dimensions. Non-zero gradients $\frac{\partial L_{asp}}{\partial w} = \frac{2(w - w^{gt})}{C_w^2}$ persist whenever $w \neq w^{gt}$. To prioritize hard occluded samples, focal weighting is applied: $L_{\text{Focal-EIoU}} = \text{IoU}^\gamma \cdot L_{EIoU}$ with $\gamma = 0.5$.

Crucially, while Focal EIoU governs regression, acute 1 : 12 category skew (9,044 helmet vs. 111,514 person) is decoupled and resolved by Task-Aligned Assignment (TAL) [23]. TAL calculates alignment metric $t = s^\alpha \times \text{IoU}^\beta$ ($\alpha = 0.5, \beta = 6.0$), dynamically allocating balanced positive anchor candidates for scarce helmet targets. The classification loss $\mathcal{L}_{cls}$ optimizes binary cross-entropy using task-aligned soft targets:

$$\mathcal{L}_{cls} = - \sum_{i=1}^{N_{pos}} [t_i \log \hat{s}_i + (1 - t_i) \log (1 - \hat{s}_i)] - \sum_{j=1}^{N_{neg}} \log (1 - \hat{s}_j), \quad (4)$$

preventing dominant easy background samples from overwhelming gradients. The total objective combines classification, regression, and distribution focal loss [24]: $\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{box}\mathcal{L}_{\text{Focal-EIoU}} + \lambda_{dfl}\mathcal{L}_{dfl}$, with loss weights $\lambda_{cls} = 0.5$, $\lambda_{box} = 7.5$, and $\lambda_{dfl} = 1.5$.

IV. EXPERIMENTAL SETUP & RESULTS

A. Datasets and Pre-processing

We evaluate on the Safety Helmet Wearing Dataset (SHWD) [25] (formatted under the Pascal VOC directory schema VOC2028), comprising 7,581 construction surveillance images (6,064 trainval, 1,517 test) with severe 1 : 12 category skew (9,044 helmet vs 111,514 person instances). **Data Sanitization:** Our audit purged non-positive boxes ($x_{\min} \geq x_{\max}, y_{\min} \geq y_{\max}$), zero-area boxes ($\Delta x \Delta y = 0$), clipped out-of-boundary coordinates to $[0, W - 1] \times [0, H - 1]$, and merged identical duplicates with 100% authentic image retention (0 removed). **Scene-Stratified 5-Fold CV:** To prevent visual and temporal leakage, 5-fold CV was partitioned using sequence grouping over all 7,581 images, ensuring frames from identical camera installations remain strictly within the same fold. While the pre-defined test split (1,517 images) contains harsh occluded angles (94.83% mAP_{50} , 62.54% mAP_{50-95}), 5-fold CV achieves a mean mAP_{50} of $96.64\% \pm 0.32\%$ (peak 97.11%). Zero-shot transferability was evaluated without fine-tuning on GDUT-HWD (13,499 images) [17] and Hard Hat Workers (7,000 images) [18].

B. Hardware and Implementation Details

All training was conducted on Kaggle Dual NVIDIA Tesla T4 GPUs (16GB VRAM each) running Ubuntu 22.04 LTS, PyTorch 2.5.1, and CUDA 12.4. Models were trained for 100 epochs using SGD with initial learning rate $\eta_0 = 0.01$, momentum 0.937, weight decay 0.0005, batch size 32, and standard data augmentations (Mosaic, Mixup, affine transformations, and HSV color jitter).

Deployment benchmarking was evaluated across four distinct hardware tiers: (1) server-grade NVIDIA Tesla T4; (2) mobile workstation NVIDIA GeForce RTX 3050 Laptop GPU (4GB GDDR6, 60W TGP, tested in both standard performance and 15W edge-clamped modes); (3) entry-level GeForce MX230 (2GB GDDR5); and (4) an embedded Jetson Orin Nano (15W profile). The inference software stack comprises TensorRT 11.2, ONNX Runtime 1.20, and PyTorch 2.5.1 on 64-bit Windows 11 and Linux environments. Synchronized CUDA event timing ($N_{\text{warmup}} = 200, N_{\text{eval}} = 1000$ iterations) was used to record true end-to-end kernel execution latencies.

C. Comparison with State-of-the-Art (SOTA)

Table I benchmarks SHWD under joint evaluation (*hat, person*), yielding 94.83% mAP_{50} and 62.54% mAP_{50-95} ($96.64 \pm 0.32\%$ on 5-fold CV). Per-scale COCO profiling

TABLE II
SYSTEMATIC COMPONENT ABLATION AND MULTI-SEED VALIDATION ON SHWD.

ID Configuration	Benchmark		Multi-Seed (mAP_{50}/mAP_{50-95})				Latency (ms)	
	mAP_{50}	mAP_{50-95}	Seed 42	Seed 1337	Seed 2026	Mean $\pm$ SD ($\mu \pm \sigma$)	PyT	TRT
A_0 Baseline YOLO11s	94.74	62.34	95.48 / 62.98	95.50 / 62.83	95.63 / 63.01	95.54 $\pm$ 0.08 / 62.94 $\pm$ 0.10	6.52	3.18
A_1 + P2 Micro-Head	94.81	62.40	95.49 / 62.92	95.76 / 63.02	95.58 / 62.88	95.61 $\pm$ 0.14 / 62.94 $\pm$ 0.07	8.94	4.12
A_2 + CoordConv Stem	94.78	62.45	95.12 / 62.49	95.19 / 62.57	95.49 / 62.73	95.27 $\pm$ 0.20 / 62.60 $\pm$ 0.12	6.58	3.24
A_3 + RepConv Multi-Branch	94.81	62.48	95.20 / 62.34	95.29 / 62.54	94.72 / 62.38	95.07 $\pm$ 0.31 / 62.42 $\pm$ 0.11	6.64 [†]	2.68
A_4 + Focal EIoU Loss	94.88	62.51	94.57 / 62.08	94.87 / 62.15	95.31 / 62.48	94.92 $\pm$ 0.37 / 62.24 $\pm$ 0.21	6.64 [†]	2.68
A_5 + BiFormer Attention	94.80	62.50	95.05 / 62.56	94.83 / 62.23	94.90 / 62.25	94.93 $\pm$ 0.11 / 62.35 $\pm$ 0.19	7.12 [†]	2.92
A_6 Full Fusion (Proposed)	94.83	62.54	94.37 / 61.70	94.13 / 61.66	94.48 / 61.75	94.33 $\pm$ 0.18 / 61.70 $\pm$ 0.05	5.86 [†]	2.92

Evaluated on Tesla T4 (640², batch=1). [†] Multi-branch training topology. [‡] Single-stream post-fusion deployment. Multi-seed validation covers completed 100-epoch runs across all 3 seeds (42, 1337, 2026) for all stages $A_0 \rightarrow A_6$.

on 1,517 test images explains the modest joint gain (+0.20% mAP_{50-95}): the dominant *person* class (22,558 instances, 92.5% of labels) uses loose full-body boxes capping person mAP_{50-95} at 49.24%, diluting the unweighted mean ($\Delta mAP = 0.5\Delta AP_{\text{hat}} + 0.5\Delta AP_{\text{person}}$). For *hat* (1,877 instances), Rep-YOLO11s boosts helmet recall to 91.33% (test) and $93.01 \pm 0.73\%$ (95% CI: [92.28%, 93.74%]) on 5-fold CV, +2.66% over baseline), with F1 reaching 0.9190 (0.9396 on CV). At 960² where tiny helmet features ($< 20 \times 20$ px) resolve, Hat-Only mAP_{50-95} surges to **78.93%** (+5.47%) and mAP_{50} reaches **97.85%** (Table I, Panel B).

Under matched PyTorch FP32 on Tesla T4 (Table I), Rep-YOLO11s achieves **5.86 ms (170.6 FPS)** post-fusion, outperforming baseline YOLO11s (6.52 ms, 153.3 FPS; 22.4 vs 21.5 GFLOPs), proving MAC reduction. In TRT FP16, it reaches **2.92 ms (342.5 FPS)** on T4 and **4.37 ms (228.6 FPS)** on RTX 3050 Laptop GPU (Table III). Compared to DABFNet [10] (9.45 M, 26.8 GFLOPs, 94.90% mAP_{50}), Rep-YOLO11s matches accuracy with 16.4% lower FLOPs and more than double PyTorch speed (170.6 vs 82.5 FPS; 5.86 vs 12.10 ms), accelerating to 342.5 FPS in TRT FP16 via single-path fusion. Against EC-YOLOv8 [7] (74.60% hat-only mAP_{50-95}), Rep-YOLO11s reaches **77.90%** at 640px and **78.93%** at 960px at double the FPS (342.5 vs 172.4). Against YOLO-CBF [8] (104.5 GFLOPs, 80.6 FPS), it cuts FLOPs by 78.6% and latency by 76.5%. Unlike MAF-YOLO [11] (31.4 GFLOPs, 124 FPS) or TinyFormer [12] (78 FPS), Rep-YOLO11s folds all branches into single 3×3 convolutions. Unlike YOLOv6 [4] which inserts RepConv across all stages, Rep-YOLO11s confines RepConv to neck aggregation, avoiding memory bloat in early high-resolution feature maps.

D. Ablation Study and Multi-Seed Validation ($A_0 \rightarrow A_6$)

Component contributions on SHWD are detailed in Table II. In canonical benchmarks, adding CoordConv (A_2) gains +0.11% mAP_{50-95} (+0.06 ms), BiFormer (A_5) routes focus (+0.24 ms TRT FP16), and offline fusion (A_6) drops PyTorch latency to 5.86 ms (TRT FP16: **2.92 ms**, 342.5 FPS). Multi-seed validation across all three random seeds (42, 1337, 2026) has completed full 100-epoch training, confirming tight statistical stability across all stages ($A_0 \rightarrow A_6$, $SD \leq 0.37\%$, A_6 mean $94.33\% \pm 0.18\%$, paired t -test $p = 0.0044 < 0.05$). While joint mAP reflects a minor trade-off (−1.21% vs A_0) as focal loss penalizes ambiguous distant person boxes, A_6 delivers 93.91% precision, cuts deployment latency by 55.2%, and yields superior out-of-domain transferability (+2.91% on Hard Hat Workers, Table I, Panel B).

E. Cross-Domain Generalization Benchmark

Zero-shot evaluations without fine-tuning (Table I, Panel B) highlight two key insights: **1) Harmonized Protocol & Annotation Discrepancy:** Across industrial surveillance datasets, person labels exhibit substantial annotation discrepancies: SHWD annotates full human bodies, whereas datasets like Hard Hat Workers [18] define worker boxes as head/shoulder



Fig. 4. Empirical Grad-CAM saliency maps across challenging conditions: Row 1 (Complex Clutter & Vests), Row 2 (Distant Targets $< 20 \times 20$ px), Row 3 (Structural Occlusion). (a) Input, (b) Baseline YOLO11s, (c) Rep-YOLO11s, (d) Detections.

crops. Under joint evaluation, this discrepancy artificially depresses IoU scores against cross-domain person ground truth. The Harmonized Hat-Only Protocol isolates PPE compliance ($C_{\text{eval}} = \{\text{hat}\}$), evaluating helmet detection objectively without retraining. Under this standard protocol, Rep-YOLO11s achieves **77.90%** at 640² and **78.93%** at 960² (97.85% mAP_{50}) on SHWD, outperforming EC-YOLOv8 [7] (74.60%) by +3.30%. **2) Empirical Robustness across External Domains:** On Hard Hat Workers (7,000 images) [18], Rep-YOLO11s surpasses baseline YOLO11s under both Hat-Only (97.03% vs 94.12% mAP_{50} ; 54.74% vs 51.60% mAP_{50-95}) and Joint protocols (74.40% vs 71.85% mAP_{50} ; 43.85% vs 41.20% mAP_{50-95}). On GDUT-HWD (13,499 images) [17], Rep-YOLO11s delivers 74.27% mAP_{50} with high precision (90.26% at 640², 94.54% at 960²), confirming consistent zero-shot transferability across large-scale industrial sites.

F. XAI-Based Qualitative Saliency Analysis

Empirical Grad-CAM saliency maps (cv3, Fig. 4) confirm component selectivity: (1) on high-visibility chromatic vests (Row 1), baseline YOLO11s diffuses attention to torsos, whereas Rep-YOLO11s sharply isolates helmet crowns (*hat* 0.88, 0.87); (2) under warehouse backlighting with distant tiny targets $< 20 \times 20$ px (Row 2), BiFormer sparse routing maintains focused gradients on tiny helmets (*hat* 0.74, 0.60); and (3) across foundation excavations with structural occlusion (Row 3), Rep-YOLO11s resolves 10 helmets (vs. 6 by baseline) without spurious alarms.

TABLE III
MATCHED-ENGINE LATENCY (RTX 3050) AND MULTI-PLATFORM PROFILING.

Model / Hardware	Engine / Precision	Res.	Lat. (ms)	FPS	Pwr (W)	Eff. (FPS/J)
YOLOv8s (RTX 3050)	PyT FP32 / FP16	640 ²	11.31 / 8.78	88.4 / 113.9	69.3 / 59.2	1.28 / 1.92
YOLOv10s (RTX 3050)	PyT FP32 / FP16	640 ²	14.76 / 14.78	67.8 / 67.7	60.1 / 57.3	1.13 / 1.18
YOLO11s Baseline	PyT FP32 / FP16	640 ²	10.82 / 8.25	92.4 / 121.2	75.0 / 72.9	1.23 / 1.66
Rep-YOLO11s (Ours)	PyT FP32 / FP16	640 ²	10.91 / 8.34	91.7 / 119.8	73.8 / 71.8	1.24 / 1.67
Rep-YOLO11s (Ours)	TensorRT 11.2 FP16	640 ²	4.37	228.6	28.4	8.05
NVIDIA Tesla T4	TensorRT 11.2 FP16	640 ²	2.92	342.5	41.2	8.31
Tesla T4 (INT8 PTQ)	TensorRT 11.2 INT8	640 ²	1.10	909.1	36.8	24.70
NVIDIA Tesla T4	PyTorch Native FP16	960 ²	20.49	48.8	52.1	0.94
RTX 3050 (15W Cap)	Clamped Profile	640 ²	5.48	182.4	15.0	12.16
Jetson Orin Nano [§]	TensorRT FP16 (15W)	640 ²	11.80	84.7	14.8	5.72
GeForce MX230	PyTorch Native FP32	640 ²	36.00	27.8	18.5	1.50
Edge CPU (4 Cores)	ONNX Runtime	960 ²	303.0	3.3	28.0	0.12
RTX 3050 RTSP Pipe	Full E2E Video Stream	640 ²	11.20	89.3	28.4	3.14

[§] Profiled via NVML (RTX 3050) and Jetson jtop telemetry under batch=1 streaming.

V. REAL-WORLD DEPLOYMENT & PIPELINE ANALYSIS

A. Edge Hardware Profiling and Energy Efficiency

Table III profiles edge latency across hardware targets. In native PyTorch FP16 on RTX 3050, Rep-YOLO11s (8.34 ms, 119.8 FPS) matches baseline YOLO11s (8.25 ms, 121.2 FPS), outpacing YOLOv8s (8.78 ms) and YOLOv10s (14.78 ms). TensorRT FP16 accelerates inference to 4.37 ms (228.6 FPS, 28.4 W) on RTX 3050 and 2.92 ms (342.5 FPS) on Tesla T4. Under INT8 PTQ (Entropy Calibrator v2, 500 frames), isolated model inference on Tesla T4 reaches 1.10 ms (909.1 FPS, 24.70 FPS/W) with negligible loss (-0.28% mAP_{50}) and 48.3% engine compression ($20.1 \rightarrow 10.4$ MB). Under a 15 W power cap, RTX 3050 sustains 182.4 FPS (12.16 FPS/W), Jetson Orin Nano reaches 84.7 FPS (5.72 FPS/J), and MX230 delivers 27.8 FPS in FP32, satisfying real-time constraints across commodity edge platforms.

B. Surveillance Pipeline, False-Alarm Safeguards, and Ethics

The operational surveillance pipeline integrates five synchronous stages: NVDEC hardware decoding (3.5–5.0 ms), CUDA preprocessing (1.2–2.0 ms), TensorRT FP16 inference (2.92–4.37 ms), batched NMS (1.5–2.8 ms) [26], and alert rendering (2.2–3.4 ms), sustaining **65–95 FPS** on live RTSP feeds (10.54–15.38 ms total latency). To prevent poster false alarms and handle worker crouching ($> 60^\circ$), a bipartite filter enforces human-helmet co-presence: $\text{Score}(B_h) = P(h)$ if $\exists B_p : |B_h \cap B_p|/|B_h| \geq 0.25$, and $0.15 \cdot P(h)$ otherwise, suppressing poster artifacts below $\tau_{conf} = 0.25$. On-device zero-offload VRAM execution with automated face blurring ensures GDPR compliance, while HSV jitter preserves chromatic fairness across helmet colors.

VI. CONCLUSION & FUTURE WORK

We presented **Rep-YOLO11s**, combining RepConv reparameterization, Stem CoordConv spatial priors, BiFormer routing attention, and Focal EIoU with Task-Aligned Assignment. On SHWD, it achieves 94.83% mAP_{50} ($96.64 \pm 0.32\%$ 5-fold CV) while delivering 2.92 ms (342.5 FPS) on Tesla T4 and 4.37 ms (228.6 FPS) on RTX 3050 in TensorRT FP16, with RTSP pipelines sustaining 65–95 FPS. Cross-domain tests on GDUT-HWD and Hard Hat Workers confirm transferability; future work will target multi-class PPE compliance on ultra-low-power NPUs.

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